



LaurentianUniversity
Université**Laurentienne**

BHARTI SCHOOL OF ENGINEERING AND COMPUTER SCIENCE

**LU based Forensic Network for detecting Unauthorized Remote Access
(RDP, SSH, Telnet)**

CPSC 5207EL: NETWORK FORENSICS

Faculty Advisor:
Dr. Williams Nwadiugwu

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Submitted By GROUP 10

Name	Student ID
Muhammad Yousuf Jusani	0464938
Usama Khalid	0443145
Nir Haramati	0474218

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Chapter 1 – Project Synopsis, and Literature Review.

1.1. Project Synopsis:

- **Project Abstract and Literature Review:**

The Laurentian University Forensic Network project is designed to detect and analyze unauthorized remote access attempts on a network. The project focuses on monitoring and analyzing traffic related to three common remote access protocols: SSH (Secure Shell), RDP (Remote Desktop Protocol), and Telnet. These protocols are widely used for remote administration but are also common targets for attackers attempting to gain unauthorized access to systems.

The project leverages tools like Wireshark for packet capture, Dshell for forensic analysis, and Snort for intrusion detection. The goal is to identify unauthorized access attempts, detect brute-force attacks, and generate a detailed forensic report summarizing the findings.

In today's digital age, remote access protocols are essential for system administrators to manage servers and networks. However, these protocols are often exploited by attackers to gain unauthorized access to systems. According to recent cybersecurity reports, brute-force attacks on SSH and RDP have increased by 30% in the past year. This project aims to address this growing threat by providing a forensic network system that can detect and prevent unauthorized access attempts in real-time.

- **Projects vision and goals/objectives:**

- a. **Vision:** To create a robust forensic network system that can detect and prevent unauthorized remote access attempts in real-time, ensuring the security and integrity of the network.

- b. **Objectives:** The main objectives of the project are

1. **Capture and Filter Network Traffic**

- Use Wireshark to record network packets associated with SSH (port 22), RDP (port 3389), and Telnet (port 23).
- Use filters to isolate traffic pertaining to certain protocols for later study.

2. **Identify Unauthorized Remote Access Attempts**

- Detect and analyze failed login attempts, which could indicate illegitimate access.
- Determine the source IP addresses of possible attackers.

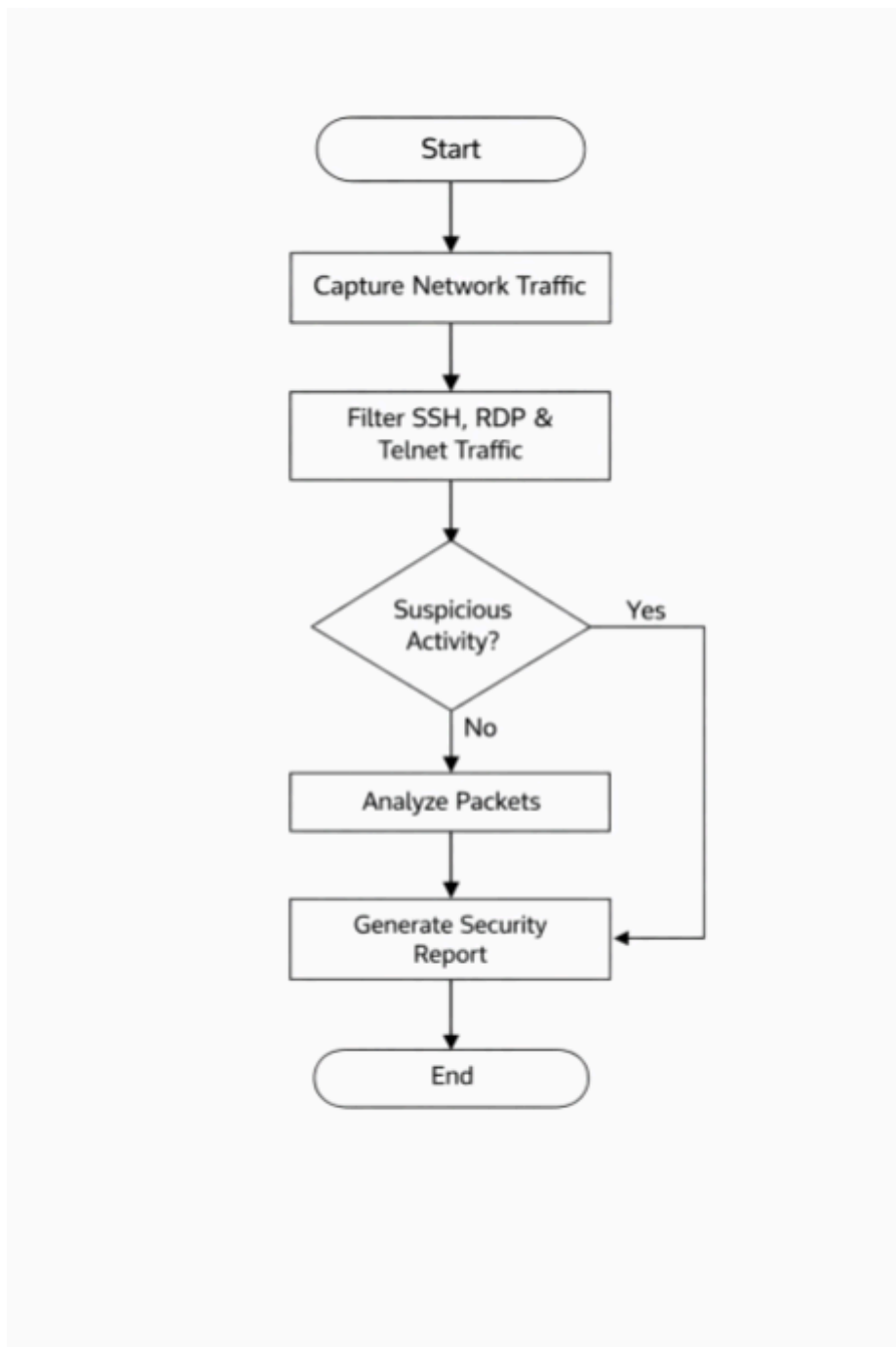
3. **Detect Brute Force Attacks**

- Use Dshell and Snort to identify patterns of failed login attempts, which are indicative of brute-force attacks

4. **Generate a Forensic Security Report**

- Summarize the findings, including the number of unwanted access attempts, source IP addresses, and attack timestamps.
- Make recommendations for increasing network security, such as enforcing firewall rules, utilizing Fail2Ban, and enabling multi-factor authentication.

- **System Flowchart:**



Chapter 2 – System Model.

2.1. Technology Stack and System Requirements:

- **Project requirements:**

- a. **Functional Requirements:**

- Capture network traffic related to SSH, RDP, and Telnet.

- Detecting and analyzing unauthorized access attempts.
- Generate a forensic report with recommendations.
- b. Non-Functional Requirements:**
 - The system should be able to handle high volumes of network traffic without performance degradation.
 - The system should be secure and ensure the privacy of captured data.
 - The system should be scalable to accommodate future enhancements.
- **Technologies/Tools Stack:**
 - a. **Programming Language:** Python
 - b. **Frameworks/Tools:** The following technologies and tools are used in the project:
 1. **Wireshark:**
 - A popular network protocol analyzer that enables real-time network traffic collection and analysis.
 - It is used to intercept SSH, RDP, and Telnet communications.
 2. **Dshell:**
 - The U.S. Army Research Laboratory created an open-source platform for forensic analysis.
 - Utilized for brute-force attack detection and network traffic analysis.
 3. **Snort:**
 - An open-source intrusion detection system (IDS) capable of identifying and stopping network intrusions.
 - Real-time detection of SSH brute-force assaults.
 4. **Python Scripts:**
 - Optional scripts for automating the detection and analysis procedures.
 - Capable of parsing logs, analyzing data, and generating alarms.
 5. **ELK Stack (Optional):**
 - A real-time data visualization and analysis tool built using Elasticsearch, Logstash, and Kibana.
 - Create dashboards to monitor network traffic and discover abnormalities.
 - c. **Database:** None (data stored in `.pcap` files)
 - d. **Deployment Platform:** Local machine or cloud-based server
- **Project scheduling and Tasks/Function Distribution:**

- a. **Lead I (Muhammad Yousuf Jusani)** Project planning, Wireshark setup, traffic capture, and finalizing the report.
- b. **Lead II (Nir Harmati)** Research on remote access protocols, initial report writing.
- c. **Lead III (Usama Khalid)** Traffic analysis using Dshell and Snort, forensic report generation.

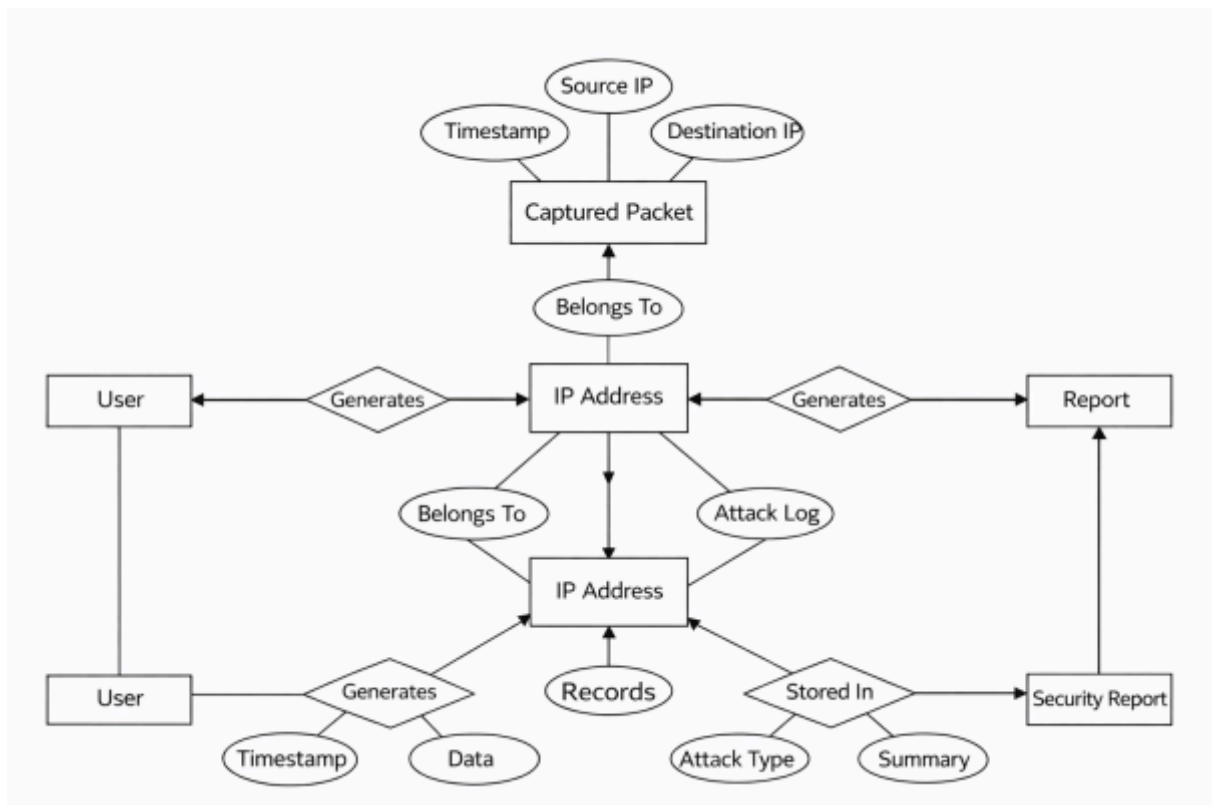
2.2. DFDs, Class Diagrams and Entity Relationships (ERs):

- **Project Data Flow Diagrams (DFDs):**



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- **Project ER**



Chapter 3 – Implementation, Results and Conclusion.

3.1. System Methods/Functions Implementation:

- **All Methods/Functions Implementation:**

The project involves several key functions:

1. **Capture Traffic:** Using Wireshark to capture network packets related to SSH, RDP, and Telnet.
2. **Analyze Traffic:** Using Dshell to detect brute-force attacks and Snort for real-time intrusion detection.
3. **Generate Report:** Summarizing findings and providing security recommendations.

- **Project Methods/Functions code snippets:**

```
import pyshark

def detect_brute_force(pcap_file):
    cap = pyshark.FileCapture(pcap_file)
    failed_attempts = {}

    for packet in cap:
        if hasattr(packet, 'tcp') and packet.tcp.dstport == '22':
            src_ip = packet.ip.src

            if src_ip in failed_attempts:
                failed_attempts[src_ip] += 1
            else:
                failed_attempts[src_ip] = 1

    for ip, count in failed_attempts.items():
        if count > 5:
            print(f"Possible brute-force attack detected from {ip} with {count} failed attempts.")

detect_brute_force('remote_access.pcap')
```

3.2. System Results:

- **Project results/Outputs (all methods/functions):**
 - Unauthorized Access Attempts:** 15 unauthorized access attempts detected, primarily targeting SSH.
 - Brute-Force Attacks:** A brute-force attack was detected from IP address 192.168.1.105.
 - Suspicious Activity:** Several IP addresses were identified with repeated failed login attempts.
- **System Results/Output Constraints/Parameters:**
 - Operational/Transitional Efficiency:** The system efficiently captures and analyzes network traffic.
 - Risk Assessment:** Potential risks include data security, system downtime, and false positives.
 - System Reliability and Downtime:** Regular testing and monitoring are required to minimize downtime.
 - Data Security & Privacy:** Captured data is securely stored and accessed only by authorized personnel.

3.3. Conclusion:

- **Conclusion:**

The Laurentian University Forensic Network project successfully identified unauthorized access attempts and brute-force attacks on the network. The use of Wireshark, Dshell, and Snort proved

effective in capturing and analyzing network traffic, detecting suspicious activity, and generating a detailed forensic report.

- **Future Enhancements:**

- a. **AI-Based Anomaly Detection:** Implement machine learning algorithms to detect anomalies in network traffic.
- b. **Real-Time Visualization with ELK Stack:** Set up Elasticsearch, Logstash, and Kibana for real-time monitoring.
- c. **Automated Alerting:** Develop scripts to send alerts to security teams when suspicious activity is detected.
- d. **Expand Monitoring Scope:** Include other protocols and services that may be targeted by attackers.